

DTSC 5930

Professional Internship

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	None
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Machine Learning Engineer (48.9 / 100)	38.2 / 100

38

CURRENT AVG SCORE
across 6 job roles

88

POTENTIAL SCORE
with recommended additions

+68

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Academic credit structure for internships with industry partners, including mid-term and final performance assessments.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	48.9	95.0	+46.1	HIGH
AI Engineer	45.4	99.3	+54.0	HIGH
ML Engineer	43.9	63.8	+19.9	MEDIUM
Applied Scientist	35.5	83.6	+48.2	HIGH
Data Scientist	34.5	93.9	+59.4	HIGH
MLOps Engineer	20.9	89.4	+68.5	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Model Deployment & Serving 2 of 6 roles

production

Experiment Tracking 1 of 6 roles

tracking

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 HIGH Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2 MEDIUM Consider Model Deployment & Serving content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.
- 3 LOW Optional: Experiment Tracking** — would broaden appeal to one additional role. Candidate terms: tracking.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

48.9 → 95.0 (+46.1 pts)

data web engineering sets learning solutions design science

AI Engineer

45.4 → 99.3 (+54.0 pts)

following data engineering models learning software code model

ML Engineer

43.9 → 63.8 (+19.9 pts)

data learning machine systems models model production engineering

Applied Scientist

35.5 → 83.6 (+48.2 pts)

biology computational problems models data learning systems engineering

Data Scientist

34.5 → 93.9 (+59.4 pts)

data sets characteristics scripting social engineering people problems

MLOps Engineer

20.9 → 89.4 (+68.5 pts)

data model production scientists systems learning machine tracking

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.

